An inference engine for eForth

version 1.0 - Thursday 15 October 2026



Marc PETREMANN

Table of Contents

Introduction **3**

The development environment **4**

The INFERENCE.FS file 4

The main.fs project 4

The other files 4

Introduction

An **inference engine** is the central component of an expert system or symbolic artificial intelligence, responsible for automated decision-making and reasoning. It works by combining a **knowledge base** (made up of facts and production rules) with deductive algorithms to derive new information.

He primarily uses two reasoning strategies:

- **The chaining before** , which starts from the available facts to arrive at a conclusion.
- **The back-chaining** , which starts from a target hypothesis to work back to the necessary facts that validate it.

In short, it simulates human logic to solve complex problems without requiring rigid procedural programming.

The development environment

For the inference engine, we will use the eForth Windows version. It's a FORTH version usable for Windows, available here:

https://github.com/MPETREMANN11/eForth-Windows/tree/main/__program

The documentation for this version is available here:

https://github.com/MPETREMANN11/eForth-Windows/tree/main/__documentation

eForth requires no installation.

Simply download the **uEf64-7.0.7.21.exe file** into a directory, in the location of your choice, named for example **eForth** .

eForth directory . Create a new folder named **INFERENCE** . Download the necessary files from the software repository. You should have:

```
📁 eForth
    uEf64-7.0.7.21.exe
    INFERENCE.fs
📁 INFERENCE
    config.fs
    facts.fs
    inference-engine.fs
    main.fs
    rules.fs
    tests.fs
```

The INFERENCE.FS file

This is the file that allows you to connect to the rest of the project. Contents:

```
\ load INFERENCE
s" INFERENCE/main.fs" included
```

Each time you want to compile the code and use the inference engine, simply launch **eForth** by clicking on the **uEf64-7.0.7.21.exe file** . Once eForth has started, type:

```
include INFERENCE.fs
```

The main.fs project

The entire project is compiled from **main.fs.** Example content:

```
include config.fs\load config
include inference-engine.fs\load inference engine
```

You can add the loading of other files. For example, to compile **tests.fs** :

```
include config.fs \ load config
include inference-engine.fs \ load inference engine
include tests.fs
```

The other files

They are loaded from **main.fs.** The project is divided into functionalities that can be used as the project evolves. The list and content of the source files may change without notice and may no longer be those described in this document.

facts.fs cannot be loaded before the contents of **inference-engine.fs** have been processed .